STAC67: Regression Analysis

Lecture 18

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Review of Wednesday

- Do the groups share one regression function? An F test compares one regression for all the groups with a separate regression in each group (the indicators and all their products with the predictors).
- Polynomial regression is multiple regression, with powers and products as extra columns of $\boldsymbol{X}$. Centre the predictors, prefer low orders, do not extrapolate.
- A product $X_1 X_2$: the slope of one predictor changes linearly with the other.

Example: power cells

A preliminary experiment on a new type of power cell. Response Y: the number of discharge-charge cycles before the cell failed. Predictors: the charge rate X_1 (amperes) and the temperature X_2 (degrees Celsius), each at three levels.

Y	150	86	49	288	157	131	184	109	279	235	224
X_1	0.6	1.0	1.4	0.6	1.0	1.0	1.0	1.4	0.6	1.0	1.4
X_2	10	10	10	20	20	20	20	20	30	30	30

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Each of the $3 \times 3 = 9$ combinations of a charge rate and a temperature was tested on one cell, except the middle one, $(X_1, X_2) = (1.0, 20)$, which was tested on three cells: $n = 9 + 2 = 11$.

The second-order model has $p = 6$ parameters, so $n - p = 5$ degrees of freedom are left to estimate σ^2.

The second-order model on the raw predictors

`lm(Y ~ X1 + X2 + I(X1^2) + I(X2^2) + I(X1 * X2))`:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	337.7215	149.9616	2.2521	0.0741
X1	-539.5175	268.8603	-2.0067	0.1011
X2	8.9171	9.1825	0.9711	0.3761
I(X1^2)	171.2171	127.1255	1.3468	0.2359
I(X2^2)	-0.1061	0.2034	-0.5214	0.6244
I(X1 * X2)	2.8750	4.0468	0.7104	0.5092

$R^2 = 0.914$, $F^* = 10.57$ on 5 and 5 df, $p = 0.011$.

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Indeed $r(X_1, X_1^2) = 0.991$ and $r(X_2, X_2^2) = 0.986$: for X_1, the factor $1/(1 - r^2)$ of Lecture 16 is already 56.

Centring, and the centred fit

Centre at the means: $x_1 = X_1 - 1$, $x_2 = X_2 - 20$. The correlation between the linear term and its square drops from 0.991 to 0.000, for both predictors. The same model on `x1` and `x2`:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	162.8421	16.6076	9.8053	0.0002
x1	-139.5833	33.0418	-4.2245	0.0083
x2	7.5500	1.3217	5.7125	0.0023
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- The second-order coefficients and their tests, the fitted values and $R^2 = 0.914$ are unchanged.

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Are the three second-order terms needed at all? Need a test with full and reduced models ($F^* = 0.78$ on 3 and 5 df).

Transforming the response

Reading a model with a log response

Survival time Y (days) of 54 patients after a liver operation, on four scores. As in Lecture 8 and Assignment 2, Box-Cox gives $\hat{\lambda} = 0.03$, with 95% interval $(-0.25, 0.30)$: take $\ln Y$. `lm(log(Y) ~ X1 + X2 + X3 + X4)`:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.8519	0.2663	14.4666	0.0000
X1	0.0837	0.0288	2.9042	0.0055
X2	0.0127	0.0023	5.4736	0.0000
X3	0.0156	0.0021	7.4403	0.0000
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One more unit of X_k, the others held fixed, adds b_k to $\ln Y$: it multiplies Y by e^{b_k}, a change of $100\,(e^{b_k} - 1)\,\%$.

- Enzyme score: $e^{0.0156} = 1.016$, survival 1.6% longer per point; 95% interval from $e^{0.0114}$ to $e^{0.0198}$: 1.1% to 2.0%.
- Blood clotting score: $e^{0.0837} = 1.087$, 8.7% per point.

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What is multiplied is the median of Y (Assignment 2, Question 5): $e^{\hat{Y}'}$, with $\hat{Y}'$ the fitted value of $\ln Y$, estimates the median survival time, not the mean.

Transforming a predictor

Transform Y, or transform X?

Two different tools:

- Transforming Y changes the shape of the relationship and the distribution of the errors: their spread and their skewness (asymmetry: a longer tail on one side). It is the remedy for unequal variances and non-normal errors.
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So when the residuals have a constant spread and look normal, but curve when plotted against a predictor: transform that predictor, or add its square.

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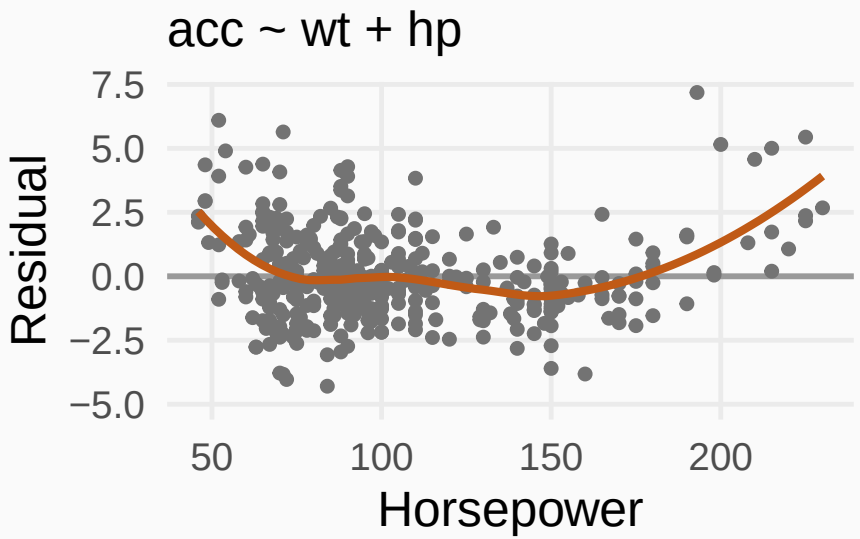
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Plot the residuals against each predictor, not only against the fitted values. Usual choices: $\ln X$, $\sqrt{X}$, $1/X$.

Example: acceleration and horsepower

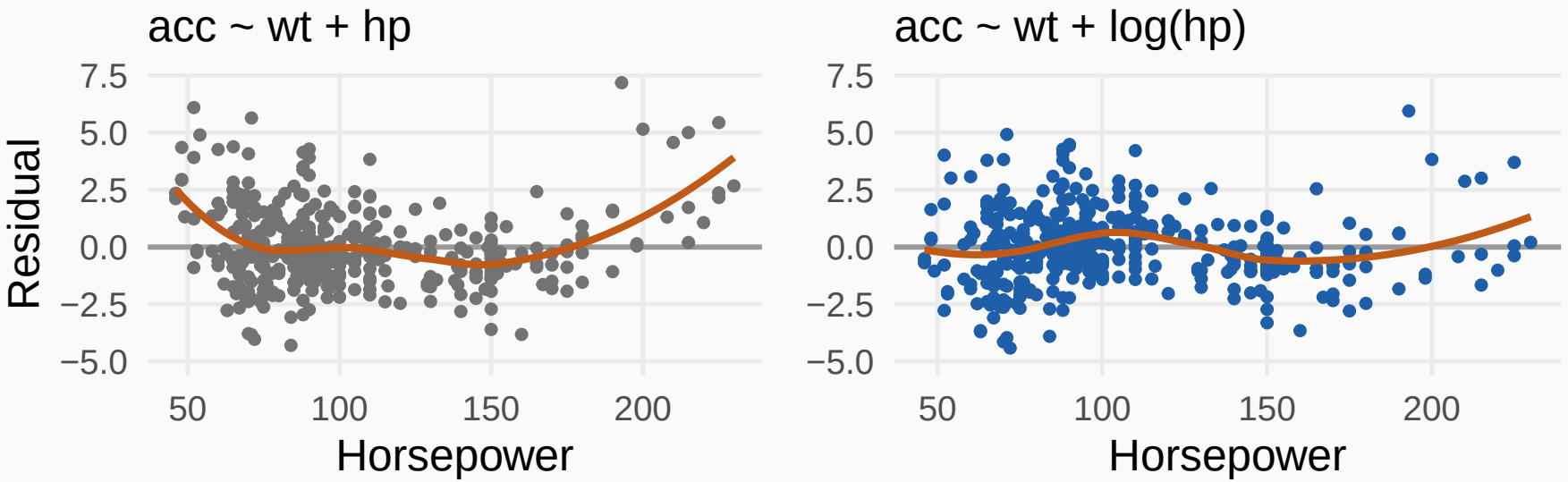
`acc ~ wt + hp`, our model since Lecture 9 (`acc`: seconds from 0 to 60 mph). Residuals against horsepower, with a lowess curve:



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With `log(hp)` in place of `hp`, most of the curve is gone, and R^2 goes from 0.604 to 0.654 with the same number of parameters.

Reading a predictor in logs

$$\widehat{\texttt{acc}} = 60.33 + 0.0027\,\texttt{wt} - 11.53\,\ln(\texttt{hp}).$$

Let b be the coefficient of $\ln X$. Multiplying X by c adds $b \ln c$ to $\hat{Y}$, whatever the starting value of X, the other predictors held fixed.

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- 1% more horsepower: $b \ln 1.01 \approx b/100 = -0.115$ s.
- 10% more: $b \ln 1.1 = -1.10$ s. Twice the horsepower: $b \ln 2 = -7.99$ s.

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The effect is the same for equal *ratios*, not for equal differences. Ten more horsepower are worth -1.78 s from 60 to 70 hp, and only -0.56 s from 200 to 210 hp: diminishing returns.

Logs on either side: how to read b

Response	Predictor	Reading of its coefficient b
Y	X	one more unit of X: Y changes by b units
$\ln Y$	X	one more unit of X: Y times e^b, about $100\,b\,\%$ more
Y	$\ln X$	1% more X: Y changes by about $b/100$ units
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The last row is a power law, $Y \propto X^b$: for instance the fuel burnt by a ship against its speed (problem set).

A log, or a square?

Wednesday's remedy for a curve was a square. Three models for the acceleration:

Model	p	R^2	adjusted R^2
`acc ~ wt + hp`	3	0.604	0.602
`acc ~ wt + log(hp)`	3	0.654	0.652
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The last two models are not nested: no F test can compare them. We need criteria to compare any two models. This is the rest of today.

Model selection

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The right balance depends on the aim: to explain (each coefficient is read, so the model must stay readable) or to predict (only the quality of $\hat{Y}$ for new cases matters).

Two steps, and the data

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- the four scores: blood clotting X_1, prognostic index X_2, enzyme X_3, liver function X_4;
- age X_5, gender X_6, and alcohol use, a factor with three levels (none, moderate, severe), through the indicators X_7 and X_8.

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We start with X_1 to X_4 only: 16 models.

Why R^2 cannot choose

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R^2_p can compare models of the same size. Across sizes, plot the best R^2_p of each size against p: where the curve flattens, more predictors add little. The other criteria make this precise: each extra parameter has a price, and a larger model wins only if it improves the fit by more than that price.

Adjusted R^2

$$R^2_{a,p} = 1 - \frac{n-1}{n-p}\frac{SSE_p}{SST} = 1 - \frac{MSE_p}{SST/(n-1)}.$$

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How high is the price? Adding one predictor raises R^2_a exactly when its t statistic has $|t^*| > 1$ (next week's problem set). This is a low bar: adjusted R^2 tends to keep too many predictors.

Mallows' C_p: the idea

Let $\mu_i = E(Y_i)$ be the true mean of case i. The fitted value $\hat{Y}_i$ of a candidate model can miss it in two ways:

$$E\left(\hat{Y}_i - \mu_i\right)^2 = \underbrace{\left(E(\hat{Y}_i) - \mu_i\right)^2}_{\text{bias}_i^2\text{: predictors left out}} + \underbrace{Var(\hat{Y}_i)}_{\text{parameters estimated}}$$

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$$\Gamma_p = \frac{1}{\sigma^2}\Big[\sum_i \text{bias}_i^2 + \sum_i Var(\hat{Y}_i)\Big] = \frac{\sum_i \text{bias}_i^2}{\sigma^2} + p,$$

since $\sum_i Var(\hat{Y}_i) = \sigma^2 \sum_i h_{ii} = \sigma^2\, tr(\boldsymbol{H}_p) = p\,\sigma^2$ (Lecture 12).

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The trade-off in one line: a new predictor can only reduce the bias, and it always costs 1 (in the variance term).

Mallows' C_p: the estimate

By the mean of a quadratic form (Lecture 12), whether or not the model is right:

$$E(SSE_p) = E\big(\boldsymbol{Y}'(\boldsymbol{I} - \boldsymbol{H}_p)\boldsymbol{Y}\big) = (n-p)\,\sigma^2 + \sum_i \text{bias}_i^2.$$

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By the mean of a quadratic form (Lecture 12), whether or not the model is right:

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Hence $\Gamma_p = E(SSE_p)/\sigma^2 - (n-2p)$. Replace $E(SSE_p)$ by SSE_p, and σ^2 by the MSE of the model with all $P-1$ predictors, taken to have no bias:

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- Look for a small C_p: a small total error.
- A model with no bias has C_p close to p. A C_p far above p signals bias.
- For the model with all the predictors, $C_P = P$ exactly: it says nothing there.

AIC and BIC

Akaike's information criterion and the Bayesian information criterion (also called SBC):

$$AIC_p = n \ln\left(\frac{SSE_p}{n}\right) + 2p, \qquad BIC_p = n \ln\left(\frac{SSE_p}{n}\right) + p \ln n.$$

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- $\ln n > 2$ as soon as $n \geq 8$: BIC charges more per parameter, and chooses smaller models than AIC.
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In R, `extractAIC(fit)` gives AIC_p as above. `AIC(fit)` and `BIC(fit)` add a constant that is the same for every model: same ranking.

PRESS: predicting each case without it

Delete case i, fit the model on the other $n-1$ cases, and predict Y_i: call it $\hat{Y}_{i(i)}$. The prediction sum of squares is

$$PRESS_p = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_{i(i)}\right)^2.$$

Small is good. Each case is predicted by a model that has not seen it when fitted: it measures prediction, where SSE_p measures fit.

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No need for n fits. With h_{ii} the leverage of case i (Lectures 11 and 12),

$$Y_i - \hat{Y}_{i(i)} = \frac{e_i}{1-h_{ii}}, \qquad \text{so} \qquad PRESS_p = \sum_{i=1}^{n} \left(\frac{e_i}{1-h_{ii}}\right)^2 \geq SSE_p.$$

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A case with a high leverage pulls the fit towards itself: its residual e_i favors the model, and the division by $1-h_{ii}$ corrects for it.

The criteria at a glance

Criterion	A good model has	Price of one more parameter
R_p^2	(a large value)	none: cannot compare sizes
$R_{a,p}^2$	the largest value	one error degree of freedom
C_p	a small value, close to p	2, in units of σ^2
AIC_p	the smallest value	2
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All of them compare models for the same response on the same cases: not a model for Y with a model for $\ln Y$, and not models fitted on different rows.

Exercise

Surgical unit, $n = 54$, response $\ln Y$, for which $SST = 12.805$. Two models:

Predictors	SSE	$\ln(SSE/n)$
X_1, X_2, X_3, X_4	3.084	-2.863
X_1, X_3	5.783	-2.234

1. Compute R^2, adjusted R^2, C_p and AIC for each model. For C_p, the model with all the predictors is the model with the four scores.
2. Which model does each criterion prefer?

Exercise: solution

The MSE of the model with all the predictors is $3.084/49 = 0.0629$.

	X_1, X_2, X_3, X_4 $(p = 5)$	X_1, X_3 $(p = 3)$
R^2_p	$1 - \frac{3.084}{12.805} = 0.759$	0.548
$R^2_{a,p}$	$1 - \frac{53}{49} \times \frac{3.084}{12.805} = 0.739$	$1 - \frac{53}{51} \times \frac{5.783}{12.805} = 0.531$
C_p	$\frac{3.084}{0.0629} - (54 - 10) = 5.00$	$\frac{5.783}{0.0629} - (54 - 6) = 43.87$
AIC_p	$54 \times (-2.863) + 10 = -144.59$	$54 \times (-2.234) + 6 = -114.64$

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Every criterion prefers the four scores. The C_p of 43.9 for $p = 3$ says that leaving out X_2 and X_4 biases the fit badly.

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But is there a model between the two that does even better?

All 16 models: one or two predictors

The criteria for every subset of X_1, X_2, X_3, X_4 (continued on the next slide):

Predictors	p	R_p^2	$R_{a,p}^2$	C_p	AIC_p	BIC_p	$PRESS_p$
none	1	0.000	0.000	151.44	−75.72	−73.73	13.292
X_1	2	0.061	0.043	141.09	−77.10	−73.12	13.508
X_2	2	0.221	0.206	108.47	−87.21	−83.23	10.738
X_3	2	0.427	0.416	66.52	−103.81	−99.83	8.329
X_4	2	0.421	0.410	67.70	−103.27	−99.29	8.024
$X_1\ X_2$	3	0.263	0.234	101.94	−88.19	−82.23	11.055
$X_1\ X_3$	3	0.548	0.531	43.87	−114.64	−108.68	6.989
$X_1\ X_4$	3	0.430	0.408	67.96	−102.07	−96.10	8.472
$X_2\ X_3$	3	0.663	0.650	20.52	−130.48	−124.51	5.066
$X_2\ X_4$	3	0.483	0.463	57.17	−107.34	−101.38	7.474
$X_3\ X_4$	3	0.599	0.583	33.54	−121.09	−115.12	6.123

All 16 models: three or four predictors

Predictors	p	R_p^2	$R_{a,p}^2$	C_p	AIC_p	BIC_p	$PRESS_p$
$X_1\ X_2\ X_3$	4	0.757	0.743	3.39	−146.16	−138.21	3.914
$X_1\ X_2\ X_4$	4	0.487	0.456	58.36	−105.76	−97.81	7.902
$X_1\ X_3\ X_4$	4	0.612	0.589	32.96	−120.82	−112.87	6.209
$X_2\ X_3\ X_4$	4	0.718	0.701	11.43	−138.01	−130.05	4.598
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- Every criterion chooses X_1, X_2, X_3: the best value of each one, in colour, is on its row. Its $C_p = 3.39$ is below $p = 4$: no sign of bias.
- Adding X_4 raises R_p^2 a little, as it must, and makes every other criterion worse.

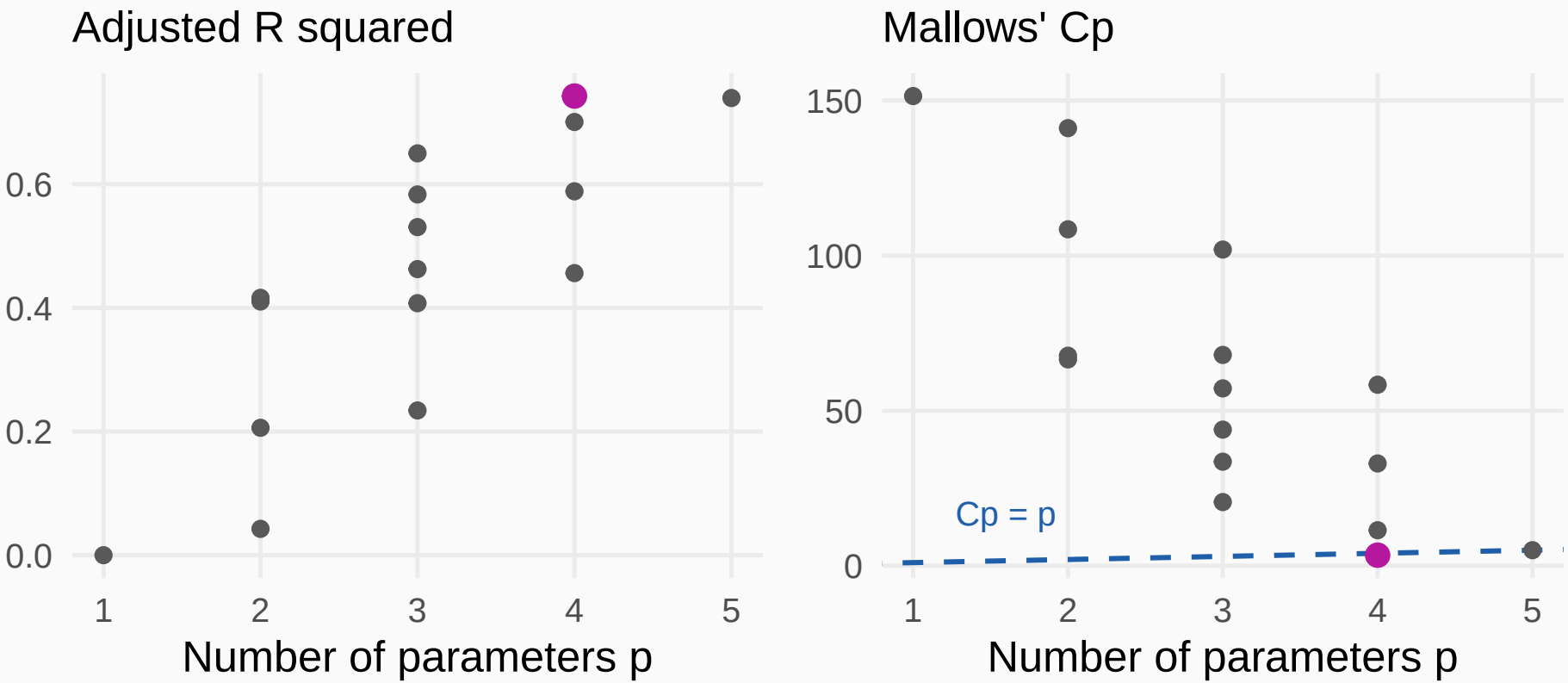
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Alone, the liver score X_4 is the second best predictor ($R^2 = 0.42$). It is correlated with the three others ($r = 0.50$, 0.37 and 0.42), and brings nothing once they are in.

Seeing the 16 models



Each point is a model; X_1, X_2, X_3 is in colour. Within a size, every criterion ranks the models as SSE_p does, except PRESS. The criteria matter when sizes are compared.

In R

For one candidate model, and the model with all the predictors:

```
fit  <- lm(log(Y) ~ X1 + X2 + X3, data = surg)
full <- lm(log(Y) ~ X1 + X2 + X3 + X4, data = surg)
n <- nrow(surg); p <- length(coef(fit))
summary(fit)$adj.r.squared                    # adjusted R2
sum(resid(fit)^2) / sigma(full)^2 - (n - 2 * p)    # Cp
extractAIC(fit)                               # p and AIC
extractAIC(fit, k = log(n))                   # p and BIC
sum((resid(fit) / (1 - hatvalues(fit)))^2)    # PRESS
```

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This gives 0.743, 3.39, -146.16, -138.21 and 3.914: the row of the table.

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`sigma(full)` is $\sqrt{MSE}$ of the full model; `hatvalues(fit)` gives the leverages h_{ii}.

With the eight predictors: 256 models

The best model of each size (smallest SSE_p), with the best value of each criterion over the 256 models in colour:

Predictors X_k	p	$R^2_{a,p}$	C_p	AIC_p	BIC_p	$PRESS_p$
3	2	0.416	117.48	−103.81	−99.83	8.329
2, 3	3	0.650	50.49	−130.48	−124.51	5.066
2, 3, 8	4	0.765	18.90	−151.00	−143.05	3.468
1, 2, 3, 8	5	0.816	5.73	−163.38	−153.43	2.736
1, 2, 3, 6, 8	6	0.821	5.53	−163.83	−151.89	2.782
1, 2, 3, 5, 6, 8	7	0.823	5.77	−163.86	−149.94	2.771
1, 2, 3, 5, 6, 7, 8	8	0.823	7.03	−162.74	−146.83	2.808
1, 2, 3, 4, 5, 6, 7, 8	9	0.819	9.00	−160.78	−142.88	2.931

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The criteria no longer agree: R^2_a and AIC keep 6 predictors, C_p keeps 5, BIC and PRESS keep 4. And the differences between these models are small.

In R: the best model of each size

The package `leaps` searches all the subsets:

```
library(leaps)
best <- regsubsets(log(Y) ~ X1 + X2 + X3 + X4 + X5 + X6 +
                     X7 + X8, data = surg, nvmax = 8)
s <- summary(best)
s$which          # the predictors of the best model of each size
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⚠ `regsubsets` sees columns, not variables. It may keep one indicator of a factor without the other (here X_8 without X_7): allowed, but X_8 then compares severe use with none and moderate together. It may also keep a square or a product without its lower-order terms, which we do not allow (Lectures 16 and 17): add those terms back before using such a model.

What the criteria do not do

- They rank models, they do not check them. The residual analysis of the chosen model is still to be done, and so is its validation.
- They often leave several models that are nearly as good. Choose among them with what you know of the subject (which predictors should matter, which are easy to measure), prefer the simpler one, and say how you chose.

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In a report, "the software chose this model" is not a justification: give the criterion, the candidates, and the reason for the final choice!

Automatic search procedures

When there are too many models

With 30 candidate predictors there are 2^{30} models, about a billion. A search procedure moves through them one predictor at a time:

- Backward elimination: start from the model with all the predictors, then remove the predictor whose removal improves the criterion most and stop when no removal improves it.
- Forward selection: start from the intercept alone, then add the predictor that improves the criterion most and stop when no addition improves it.
- Stepwise regression: forward, but after each addition, check whether a predictor already in can now be removed.

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The criterion can be a test (enter or stay if the p-value of the t test is below a threshold, often 0.10 to 0.20). R's `step()` uses the AIC.

Backward elimination with `step()`

```
full <- lm(log(Y) ~ X1 + X2 + X3 + X4 + X5 + X6 +
             X7 + X8, data = surg)
back <- step(full, direction = "backward", trace = 0)
back$anova
```

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  Step Df    Deviance Resid. Df Resid. Dev       AIC
1      NA          NA        45   1.970510 -160.7773
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`trace = 0` turns off the table that `step()` prints at each step; `back$anova` keeps the path. `Resid. Dev` is SSE_p. X_4 goes first, then X_7; any other removal would raise the AIC. The search stops at $X_1, X_2, X_3, X_5, X_6, X_8$: the smallest AIC of the table.

With `k = log(n)` in `step()`, the criterion is the BIC: two more predictors go, and it stops at X_1, X_2, X_3, X_8.

Forward selection and stepwise regression

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forw <- step(null, scope = formula(full),
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Forward selection adds $X_3, X_2, X_8, X_1, X_6, X_5$, in this order, and stops at the same model. With `direction = "both"`, the stepwise search takes the same path here.

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- A search follows one path: it can miss the best model, and the three procedures can disagree.
- A `factor` column enters or leaves with all its indicators. Here X_7 and X_8 are two separate columns, so X_7 could go alone: the model then compares severe use with the rest.
- It is a way to find candidates, not a way to choose for you.

Summary

- A response whose spread grows with its level, with skewed errors: try $\ln Y$. One more unit of X_k then multiplies the median of Y by e^{b_k}.
- A curve in the residuals against a predictor, with well-behaved errors: transform the predictor. With $\ln X$, the coefficient is read for a change of 1% in X.
- R_p^2 cannot compare models of different sizes. Adjusted R^2, C_p, AIC, BIC and PRESS each penalize more parameters.
- C_p estimates the total mean squared error: bias from the predictors left out, plus p.
- The criteria rank models for the same response on the same cases. They do not replace the residual analysis, the context, or validation.